

- (a) 339 (b) 349
(c) 369 (d) Data inadequate
(e) None of these
- 310.** In a division sum, the remainder was 71. With the same divisor but twice the dividend, the remainder is 43. Which one of the following is the divisor?
(a) 86 (b) 93
(c) 99 (d) 104
- 311.** When a certain positive integer P is divided by another positive integer, the remainder is r_1 . When a second positive integer Q is divided by the same integer, the remainder is r_2 and when $(P + Q)$ is divided by the same divisor, the remainder is r_3 . Then the divisor may be
(a) $r_1 r_2 r_3$ (b) $r_1 + r_2 + r_3$
(c) $r_1 - r_2 + r_3$ (d) $r_1 + r_2 - r_3$
(e) Cannot be determined
- 312.** Two numbers when divided by a certain divisor leave the remainders 4375 and 2986 respectively but when the sum of two numbers is divided by the same divisor, the remainder is 2361. The divisor in question is
(a) 4675 (b) 4900
(c) 5000 (d) None of these
- 313.** A number divided by 13 leaves a remainder 1 and if the quotient, thus obtained, is divided by 5, we get a remainder of 3. What will be the remainder if the number is divided by 65?
(a) 16 (b) 18
(c) 28 (d) 40
- 314.** The numbers 2272 and 875 are divided by a three-digit number N , giving the same remainder. The sum of the digits of N is
(a) 10 (b) 11
(c) 12 (d) 13
- 315.** A number when divided by three consecutive numbers 9, 11, 13 leaves the remainders 8, 9 and 8 respectively. If the order of divisors is reversed, the remainders will be (R.R.B., 2008)
(a) 10, 8, 9 (b) 10, 1, 6
(c) 8, 9, 8 (d) 9, 8, 8
- 316.** After the division of a number successively by 3, 4 and 7, the remainders obtained are 2, 1 and 4 respectively. What will be the remainder if 84 divides the same number?
(a) 41 (b) 53
(c) 75 (d) 80
- 317.** A number is successively divided by 8, 7 and 3 giving residues 3, 4 and 2 respectively and quotient 31. The number is
(a) 3555 (b) 5355
(c) 5535 (d) 5553
- 318.** A number when divided by 3 leaves a remainder 1. When the quotient is divided by 2, it leaves a remainder 1. What will be the remainder when the number is divided by 6?
(a) 2 (b) 3
(c) 4 (d) 5
- 319.** When the square of any odd number, greater than 1, is divided by 8, it always leaves remainder (I.A.M., 2007)
(a) 1 (b) 6
(c) 8 (d) Cannot be determined
- 320.** The numbers from 1 to 29 are written side by side as follows:
1234567891011121314.....2829
If this number is divided by 9, then what is the remainder? (M.A.T., 2006)
(a) 0 (b) 1
(c) 3 (d) None of these
- 321.** If 17^{200} is divided by 18, the remainder is
(a) 1 (b) 2
(c) 16 (d) 17
- 322.** What is the remainder when 2^{31} is divided by 5?
(a) 1 (b) 2
(c) 3 (d) 4
- 323.** Consider the following statements:
(1) $a^n + b^n$ is divisible by $a + b$ if $n = 2k + 1$, where k is a positive integer.
(2) $a^n - b^n$ is divisible by $a - b$ if $n = 2k$, where k is a positive integer.
Which of the statements given above is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
- 324.** $(7^{19} + 2)$ is divided by 6. The remainder is
(a) 1 (b) 2
(c) 3 (d) 5
- 325.** If $(10^{12} + 25)^2 - (10^{12} - 25)^2 = 10^n$, then the value of n is
(a) 5 (b) 10
(c) 14 (d) 20
- 326.** $(3^{25} + 3^{26} + 3^{27} + 3^{28})$ is divisible by
(a) 11 (b) 16
(c) 25 (d) 30
- 327.** $(4^{61} + 4^{62} + 4^{63} + 4^{64})$ is divisible by
(a) 3 (b) 11
(c) 13 (d) 17
- 328.** $(9^6 + 1)$ when divided by 8, would leave a remainder of
(a) 0 (b) 1
(c) 2 (d) 3
- 329.** If n is even, $(6^n - 1)$ is divisible by
(a) 6 (b) 30
(c) 35 (d) 37
- 330.** If $(12^n + 1)$ is divisible by 13, then n is
(a) 1 only (b) 12 only
(c) any odd integer (d) any even integer